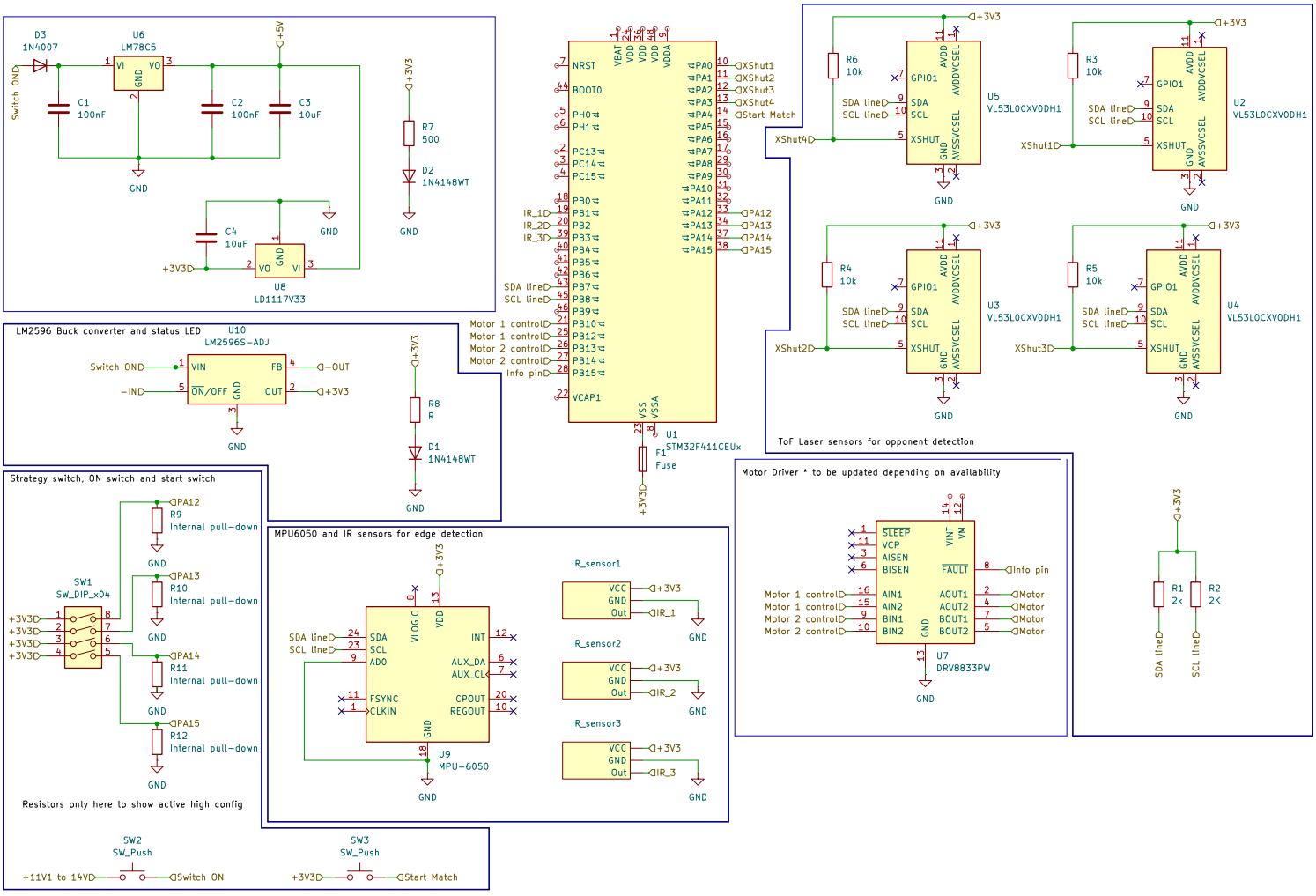

D3
1N4007
U6
LM78C5
Switch OND
C1
100nF
C2
100nF
C3
10uF
+5V
GND
+3V3
R7
500
D2
1N4148WT
C4
10uF
U8
LD1117V33
LM2596 Buck converter and status LED
U10
LM2596S-ADJ
Switch OND
-IND
VIN
FB
-OUT
+3V3
R8
R
D1
1N4148WT
U1
STM32F411CEUx
F1
Fuse
Motor 1 control
Motor 2 control
Info pin
SDA line
SCL line
IR_1
IR_2
IR_3
XShut1
XShut2
XShut3
XShut4
Start Match
ToF Laser sensors for opponent detection
R6
10k
R3
R4
R5
U5
VL53L0CXV0DH1
U2
U3
U4
GPIO1
SDA
SCL
XSHUT
AVDD
AVDDVCSEL
AVSSVCSEL
Strategy switch, ON switch and start switch
SW1
SW_DIP_x04
R9
R10
R11
R12
Internal pull-down
Resistors only here to show active high config
MPU6050 and IR sensors for edge detection
U9
MPU-6050
IR_sensor1
IR_sensor2
IR_sensor3
VCC
Out
IR_1
IR_2
IR_3
Motor Driver * to be updated depending on availability
U7
DRV8833PW
SLEEP
VCP
AISEN
BISEN
FAULT
AIN1
AIN2
BIN1
BIN2
AOUT1
AOUT2
BOUT1
BOUT2
Motor
R1
2k
R2
2K
SW2
SW_Push
+11V1 to 14V
Switch ON
SW3
SW_Push
Start Match